Final Year Project (FYP)
Week 2

Solar energy research project

- ***Project title: JOD 01: BIM to BEM: Automated Energy Model Generation Using Open Test Cases***
    - This project will investigate the automated translation of **Building Information Modelling (BIM)** data into **Building Energy Modelling (BEM)** inputs, using existing code that will be reactivated, reviewed and applied to the IEA EBC Annex 91 Subtask 2 Common Exercise. The work will focus on testing a BIM-to-BEM workflow that connects BIM authoring outputs, particularly IFC-based model files, to building performance simulation tools with minimal manual intervention. The project should begin with the simplest test case, the ASHRAE 140 Case 600 shoebox model, before extending the workflow to more complex cases, including the modified UCL Demo4 building and the Aalborg University campus models, which are intended to test geometric complexity, space-boundary handling and large-file processing. The student will reactivate and document the existing codebase, identify any required updates, and apply it to the Common Exercise stages: IFC input-file preprocessing, BEM input-file generation, and, where feasible, BEM output comparison for the ASHRAE 140 test case. The work should assess model quality, interoperability and process performance by examining issues such as geometry/topology fidelity, IFC-to-BEM mapping coverage, adjacency detection, material and construction readiness, execution time, automation share and required manual intervention. The expected outcome is a reproducible, well-documented BIM-to-BEM workflow, together with recommendations on the robustness of the existing code, the causes of data-transfer failures, and the practical steps needed to support transparent and interoperable building energy modelling workflows.
    - Contact James.ODonnell@ul.ie

CS4437 FYP deadlines

Assessment	% of Grade	Deadline
FYP Proposal Form	10%	~~28/09/2026~~ 1/10/2026
Interim FYP Report	90%	25/10/2026

Note: 30th of September session there will be no class. You can work on getting your proposal form ready and signed for submission.

FYP

- There is a synergistic benefit of tying the FYP project to your residency 4 work.

- This may not be possible due to NDAs (non-disclosure agreements).

- Also the Residency may not be finalised or perhaps a student would like to investigate their own concept or ideas that are not related to the Residency. For this reason, the FYP is not restricted to only Residency topics, but **it must be a software technology-based project**.

Non-disclosure agreement (NDA)

- All students are required to submit a FYP that doesn’t infringe on confidentiality as supervisors, second readers and externs need to be able to examine their submissions.

- It is the responsibility of each student to make sure that their FYP does not disclose any information covered by an NDA that they have signed.

Research Question

- A research question is a clear, concise, and open-ended question that centers your research.

UL Science & Engineering Ethics

- If your project does not require human participation then you do not require prior UL Science & Engineering Ethical approval.

- For any projects that do involve human participation then it is a requirement to get prior approval from the UL Science & Engineering Ethics committee.

UL writing centre

- Regional Writing Centre | University of Limerick
  - Student resources
    - Useful PowerPoints
      - Science & Engineering
        - FYP Presentation for Engineering

Academic writing in the third person

- Writing in the third person. This means not using words such as ‘I’, ‘me’ or ‘my’, and avoiding referring to yourself or your reader.

- For example, “I will be trying to find” replaced with “This report will explore”.

- Discuss with your supervisor if this is required for your FYP.

Links for literature search

- UL library: https://www.ul.ie/library/
- Journals at UL library: https://uol.primo.exlibrisgroup.com/discovery/jsearch?vid=353UOL_INST:353UOL_VU1
- All databases at UL library: https://libguides.ul.ie/az.php
- ACM digital library: https://dl.acm.org/dl.cfm
- IEEE Electronic Library online: https://ieeexplore.ieee.org/Xplore/home.jsp
- Web of Science: http://apps.webofknowledge.com/WOS_GeneralSearch_input.do;jsessionid=E7846771B92DDA437488EE26D1C22D51?product=WOS&search_mode=GeneralSearch&SID=C5nh3JV9epGLwd85bO4&preferencesSaved=

Links for literature search

- Google scholar: https://scholar.google.com/
- Scopus: https://www.scopus.com/
- https://arxiv.org/ an open access archive
- IEEE, Elsevier and Springer are three top publishers
- Check journal Impact factor
- Check conference rankings here:
  - http://portal.core.edu.au/conf-ranks/

Why impact factor and conference ranking should not be considered in isolation

- Impact Factor is a measure of the number of times an average paper in a journal is cited, during a year.
- What one field considers a high impact factor, may not be the same in another field.
- Impact factors are useful, but they should not be the only consideration when judging quality. For example, new journals won’t have an impact factor but could be good quality new research.
- Example of a A* ranked software conference: International Conference on Software Engineering (ICSE)

Using AI in your research

- If you are not knowledgeable in the area you are going to use AI in it can become a crutch instead of a tool.
- Review all AI generated content, but what else can you do?
- Your prompts need to be as detailed as possible. AI can help you write prompts.
- AI models are stateless (no memory), so, the state is in the chat. During an AI conversation the chat is used as context and passed to the AI.
- The chat has to fit in the context window, but AI models have a finite context window. Before you run out of space in your context window get a summary to use in another chat. If you run out of space in the context window, AI results will not be complete or accurate.
- Be specific with the context you give AI ,for example, tell it not to make assumptions or make up publications.

Different types of Plagiarism

- Plagiarism, Auto-Plagiarism and Self-Plagiarism
  - Academic Integrity and Academic Misconduct | University of Limerick

Library Support for Academic Writing and Project Preparation

- Academic Writing and Project/Thesis Preparation - Science & Engineering Library Research Support Guide - LibGuides at Glucksman Library University of Limerick

Referencing and citations

- Introduction to Referencing - Cite It Right: Guide to Harvard Referencing Style - LibGuides at Glucksman Library University of Limerick

- This link also has information about Reference Management Software. For example:
  - EndNote
  - Mendeley
  - Zotero

Library Sessions

- Library sessions, for example, EndNote training:
  - https://ul.libcal.com/calendars?cid=-1&t=m&d=0000-00-00&cal=-1&inc=0

EndNote

- Referencing & Endnote | University of Limerick
- https://libguides.ul.ie/referencing-endnote/desktop
- To download Endnote for free send a message to libinfo@ul.ie